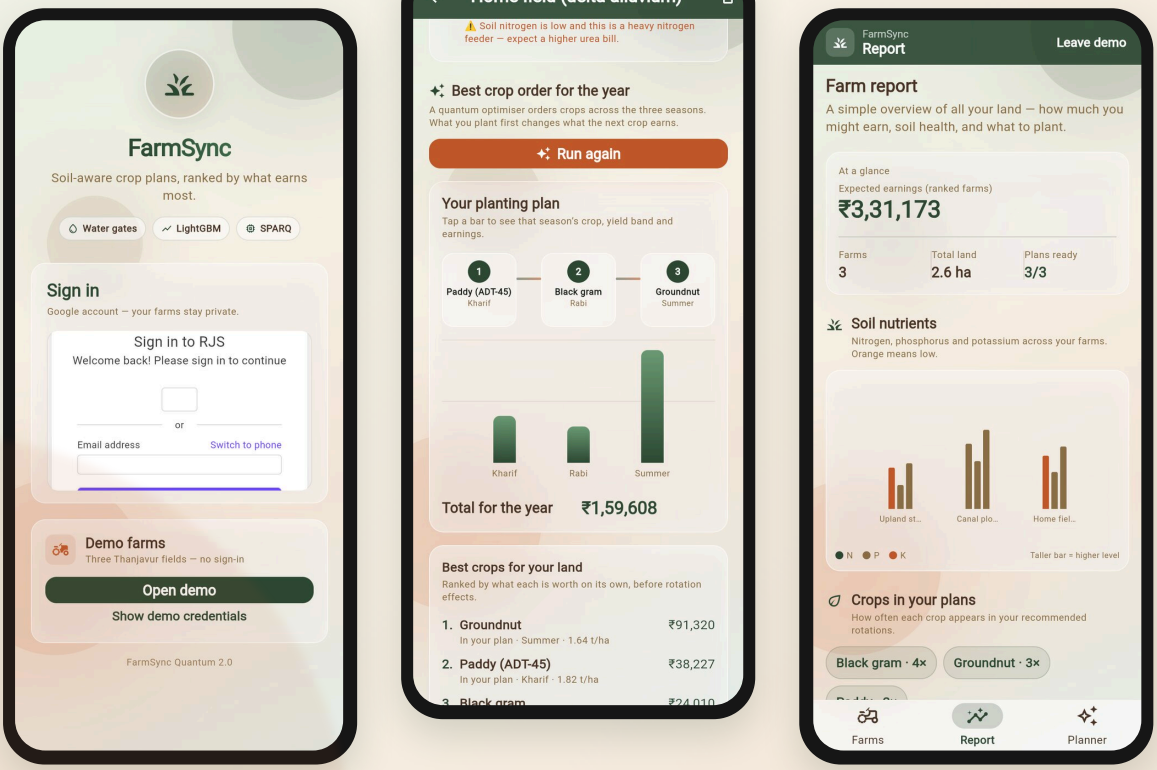
APP REPORT · OCTOBER 2026

FarmSync

What the app does, how it looks, and how well it works — explained in simple words, with real pictures from the app.



Project
High Yield Crop Prediction using Quantum AI

Prepared for
Aakash Buddha Kannan

Date
23 October 2026

Code version
main · commit 7052f6a

The short version

FarmSync is a phone app for new farmers. You tell it about your land and your soil. It tells you which crops can grow there, the best order to plant them across the year, how much money you might make, and what fertiliser your soil needs.

What the app gives a farmer

- A **soil card** that turns soil numbers into simple Low / Medium / High labels.
- A list of **crops that can grow** on that land, and a reason for each crop that cannot.
- A **crop plan for the whole year** (three seasons), picked by a quantum-style optimiser.
- **Money estimates** in rupees, plus fertiliser bags, soil fixes and water advice.
- A **"what if" planner**: drag water or budget down and watch the plan change.

How this report was made

- Read all the app and server code (216 files).
- Ran the real server on this computer and created the 3 demo farms with the real engine (live weather, trained AI model, quantum step).
- Built the real app and opened it at phone size (412 × 915). **Every app picture in this report is a real screenshot**, not a drawing.
- Ran all the automatic tests.

Our score card

Area	Rating	In one line
Idea and usefulness	Strong	Solves a real problem: what to plant, in what order, and what the soil needs.
Look and feel	Strong	Calm earthy colours, clear cards, large text, helpful charts.
Ease of use	Fair	Simple flow, but some screens are long and some words are technical. English only.
The "brain" (AI + quantum)	Strong	Works end to end. All 3 demo plans matched the best possible plan.
Reliability and testing	Needs work	Live server is offline; 24 server tests cannot run; 2 other tests fail.
Ready for real farmers?	Not yet	Great demo. Needs the fixes in section 9 before a public launch.

Top 5 things to fix first

1. **The online server is down.** The app's default server address answers "Application not found", so a phone with the app shows errors on every screen.
2. **A shared demo password is built into the app.** Anyone who installs it can open the demo account and change its farms.
3. **One test file has a typing mistake** (wrong spaces on line 392), so 24 server tests never run.
4. **The Planner starts with more water than the farm really has**, so crops that were ruled out come back.
5. **English only.** The target farmers are in Tamil Nadu; the server already stores Tamil soil names.

Contents

1 What is FarmSync?

2 How it works

3 The farmer's journey

4 Screen by screen tour

5 Look and feel (UI design)

6 Behind the scenes

7 Real results for 3 demo farms

8 Testing and health check

9 Problems found

10 What to do next

11 Word list (glossary)

A Notes on how the pictures were taken

1 What is FarmSync?

A digital guide book for farmers who are new to farming or new to their land.

The problem

A new farmer has many hard questions. Which crop will grow on my land? Is my soil good? Do I have enough water? What should I plant first, and what next? How much fertiliser do I buy? Mistakes cost money and a whole season.

The answer FarmSync gives

The farmer enters a few facts that are printed on a government **Soil Health Card** (or types them in): nitrogen, phosphorus, potassium, pH, and how much water is available. The app also knows where the farm is. From that, it builds a full plan for the year.

1. Can it grow?

Simple farm rules remove crops that cannot grow — for example, soil too salty, too little water, or clay that holds water for crops that hate wet feet.

2. How much will it earn?

An AI model, trained on 623 real records of district harvests and weather, guesses the harvest and the rupee value of each crop.

3. In what order?

A quantum-style optimiser picks the best order of crops across three seasons, because the crop you plant first changes how well the next one grows.

The three seasons used in the app

Season	When (roughly)	What it means
Kharif	June – October	The monsoon season. Rain-loving crops like paddy (rice).
Rabi	November – February	The cooler season after the rains. Pulses like black gram.
Summer	March – May	Hot and dry. Crops that need less water, like groundnut.

Who it is for

- **New and young farmers** who do not yet have years of local experience.
- **Farmers with more than one plot.** Each farm gets its own soil card and its own plan.
- **Judges, teachers and the project team**, who can see the inner workings in the "Technical details" panel and the web "Ops" page.

The app currently covers **5 crops** (paddy, black gram, groundnut, maize, sugarcane) and **5 districts**, with demo farms in **Thanjavur, Tamil Nadu**.

2 How it works

Five steps run every time a farmer taps "Rank my crops". The first four are normal computing. Only the ordering step uses the quantum method.

1 — Soil card
Soil numbers become Low / Medium / High labels using national (ICAR) limits.

2 — Crop check
Hard rules: pH, salt, water, drainage. Crops that fail are removed, with a reason.

3 — Harvest guess
AI model (LightGBM) predicts tonnes per hectare and a low–high range, using weather and soil.

4 — Best order
Quantum optimiser (SPARQ) tries crop orders for 3 seasons and keeps the best one.

5 — Advice
Fertiliser bags (urea, DAP, MOP), soil fixes and irrigation needs.

CLASSICAL COMPUTING (RULES + AI) | QUANTUM

Why the order of crops matters

If you simply plant the crops that earn the most, you can lose money. Crops affect each other:

- A **pulse** (like black gram) puts nitrogen back into the soil. The next crop needs less fertiliser.
- A **cereal after a pulse** grows better, because pests and diseases are broken up.
- The **same crop twice** in a row usually grows a bit worse.

A real example from the "Home field" demo farm

Way of choosing	Crop order (Kharif → Rabi → Summer)	Money for the year
Sort by profit (simple way)	Paddy → Groundnut → Black gram	**₹1,58,592**
FarmSync (quantum order)	**Paddy → Black gram → Groundnut**	**₹1,59,608**

Putting black gram before groundnut earns ₹1,016 more on this 1.2 ha farm, and it matched the best possible plan found by checking every option. The project's own test on 60 random farms says simple sorting gives a worse plan 65% of the time.

The quantum step in plain words

Think of 3 boxes (one per season). Each box must hold exactly one crop. The optimiser uses a small simulated quantum circuit (up to 12 "qubits") that only ever produces **valid plans — one crop per season, 100% of the time**. It runs the circuit 2,048 times, counts which plan appears most, and gives that as the answer. It runs on a normal computer as a simulation; no real quantum hardware is used.

The app is honest about this: it shows the comparison with simple sorting, and the project documents say there is no speed advantage over normal computers at this size.

3 The farmer's journey

From opening the app to having a plan takes eight simple steps.

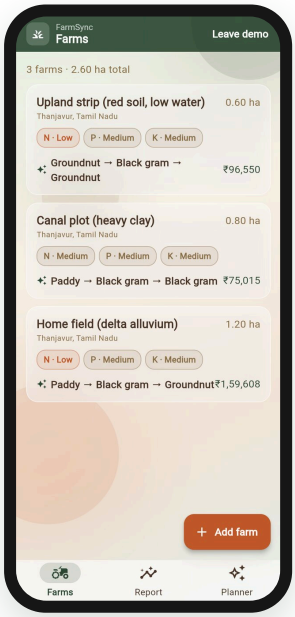
1. **Sign in** with a Google account (or open the demo farms).
2. **Add a farm:** name, land size, and location (GPS button or typed).
3. **Enter soil readings** — type them, or take a photo of the Soil Health Card and let the phone read it.
4. **See the soil card** straight away, with Low / Medium / High labels.
5. **See which crops can grow** and why the others cannot.
6. **Tap "Rank my crops"** to get the best crop order and the money estimate.
7. **Read the soil advice:** fertiliser bags, soil fix, water needs.
8. **Repeat for other farms** and see everything together in the Report tab.

The app at a glance

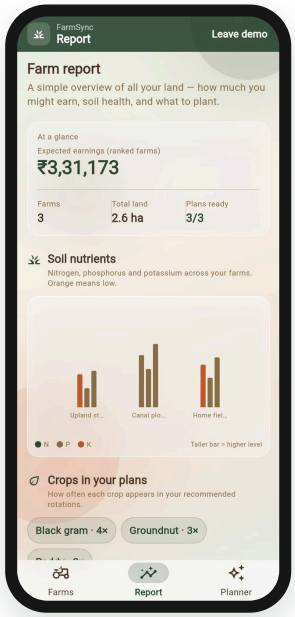
After sign-in, the app has **three tabs** at the bottom:

Tab	What it is for
Farms	The list of your farms. Add a new farm here. Tap a farm to see its full page.
Report	One summary of all farms: total expected earnings, soil chart, crops in your plans.
Planner	The "what if" tool. Move the water and budget sliders and the plan updates live.

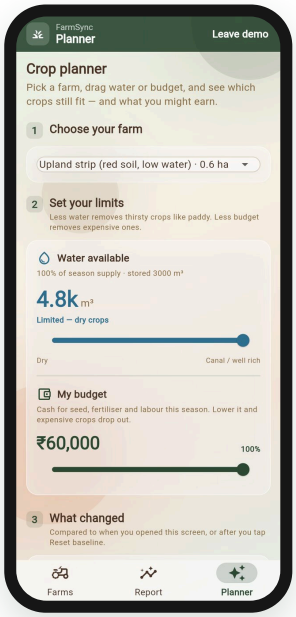
The top bar always shows the app name and the tab name. On the right is the account button (or "Leave demo" in the demo).



Farms tab



Report tab

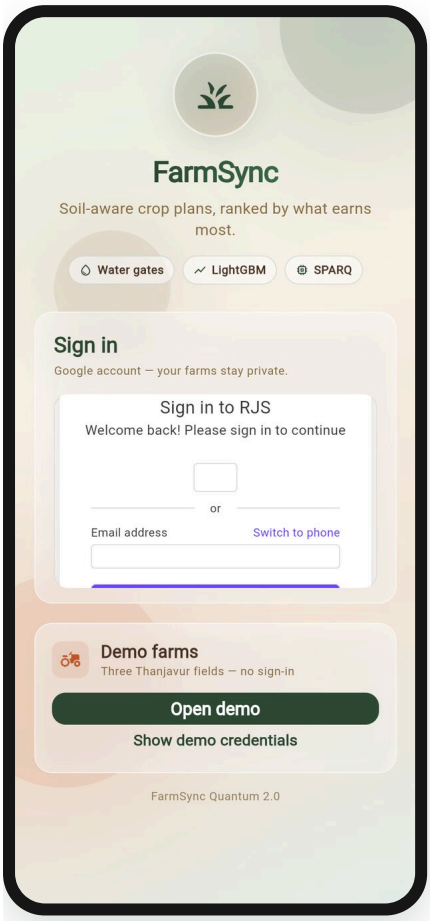


Planner tab

4 Screen by screen tour

All pictures below are real screenshots of the app, filled with the 3 demo farms.

4.1 Sign-in screen



Sign-in. Google sign-in box and demo card.

The first screen shows the FarmSync logo, the tagline *"Soil-aware crop plans, ranked by what earns most"*, and three small badges (Water gates, LightGBM, SPARQ) that hint at what is inside.

What you can do here

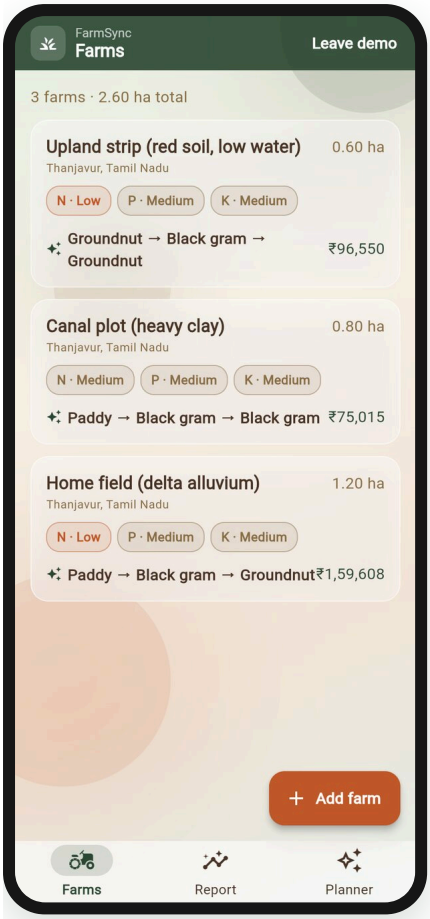
- **Sign in with Google.** This box is provided by a login service called Clerk. Your farms are private to your account.
- **Open demo.** See three ready-made farms in Thanjavur without signing in.
- **Show demo credentials.** Shows the demo account name and secret key.

What we noticed

- The login box says **"Sign in to RJS"**, not "FarmSync". The name is set in the Clerk dashboard.
- The login box is only 200 points tall, so the form is cut off and the farmer must scroll inside a small box.
- "Show demo credentials" puts a secret key on screen (see problem 2 in section 9).

Note: the Google logo did not load in our test browser (a side effect of running the phone app in a browser). On a phone it shows normally.

4.2 Farms tab



Farms list with 3 demo farms.

The top line counts the farms and total land: **"3 farms · 2.60 ha total"**. Each farm is a card that shows:

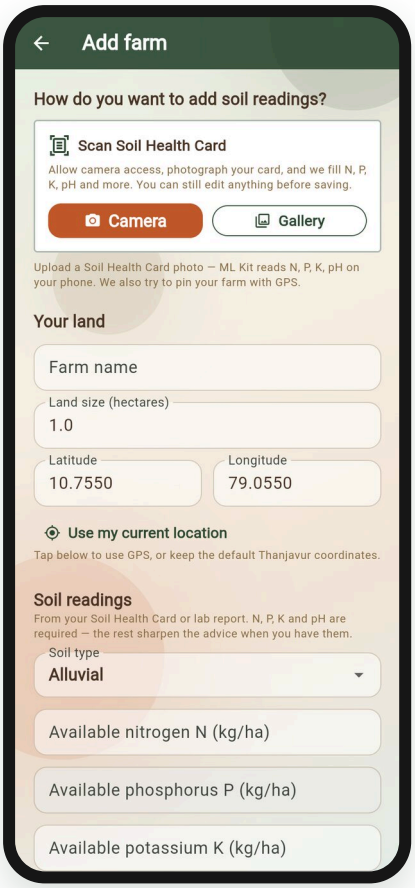
- Farm name, size in hectares, district and state.
- Coloured soil chips: **N · Low** in orange, **P · Medium** and **K · Medium** in brown. Orange always means "low — needs attention".
- The chosen crop order with a sparkle icon, for example **Groundnut → Black gram → Groundnut**, and the money estimate (₹96,550).
- If a farm has no plan yet: "Tap to rank crops" or "Add soil readings to begin".

The big orange **"+ Add farm"** button sits at the bottom right. Pull down to refresh. If there are no farms, a friendly empty screen invites you to add the first one.

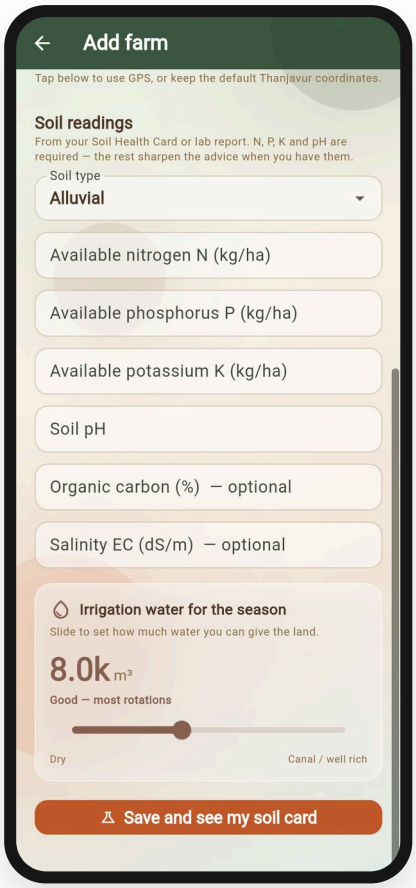
Small layout issue: on the third card the price sits right against the crop name ("Groundnut₹1,59,608") because the text is long.

4.3 Add a farm

This is a single long form with three parts: **scan the card**, **your land**, and **soil readings**.



Top of the form. Scan card, farm name, size, location.



Bottom of the form. Soil readings, water slider, save.

Scan the Soil Health Card

Tap **Camera** or **Gallery** and take a photo of the card. Google ML Kit reads the text **on the phone itself** and fills in N, P, K, pH, organic carbon, salinity and even the soil type. The app tells you which values it found and which are still missing. You can fix any number before saving.

Your land

Farm name, land size (hectares), latitude and longitude. The app tries GPS when the screen opens; there is also a **"Use my current location"** button. If GPS is not allowed, Thanjavur is used as the default place.

Soil readings

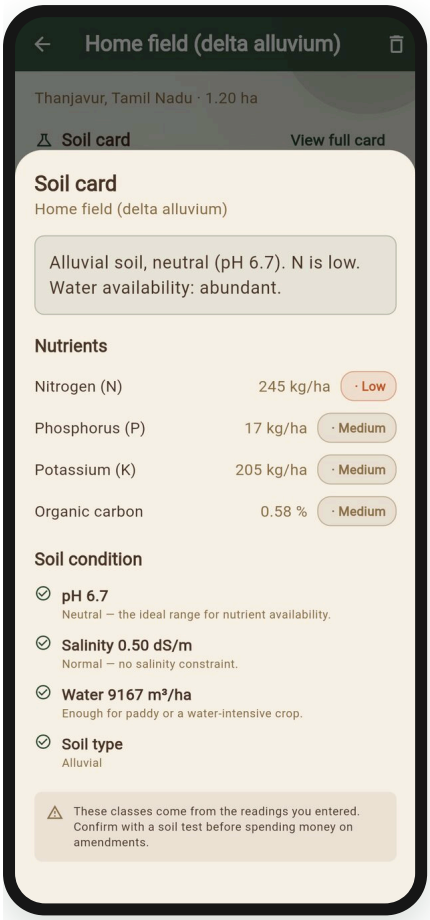
Soil type (drop-down), nitrogen, phosphorus, potassium and pH are required. Organic carbon and salinity are optional. Each box checks the number (for example, pH must be between 0 and 14).

Water slider

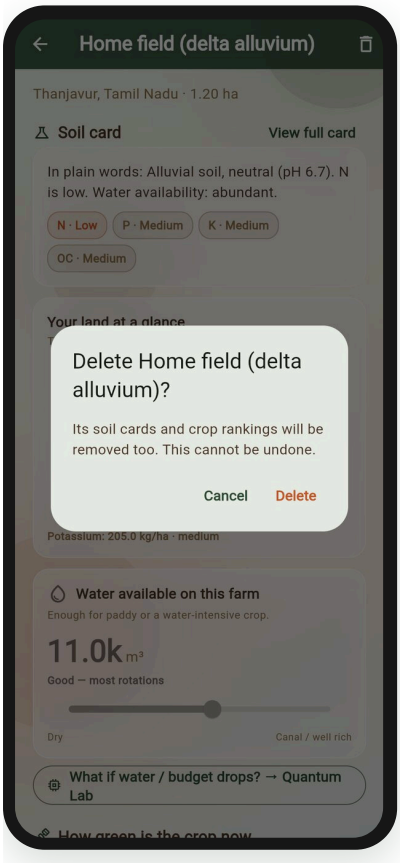
A big slider sets how much irrigation water the farm has for the season (0 – 20,000 m³). The colour moves from orange (dry) to blue (wet), with a plain label like **"Good — most rotations"**.

The orange button **"Save and see my soil card"** saves the farm and opens its soil card straight away.

4.4 Soil card



Soil card slides up from the bottom.



Delete farm asks first.

The soil card is the farmer's own numbers, explained. It opens as a sheet from the bottom of the screen.

- A green box with a one-line summary: *"Alluvial soil, neutral (pH 6.7). N is low. Water availability: abundant."*
- **Nutrients:** each value with its unit and a Low / Medium / High chip (Nitrogen 245 kg/ha · Low).
- **Soil condition:** pH, salinity, water per hectare and soil type, each with a short note such as "Neutral — the ideal range for nutrient availability".
- A clear warning at the end: *confirm with a soil test before spending money on amendments.*

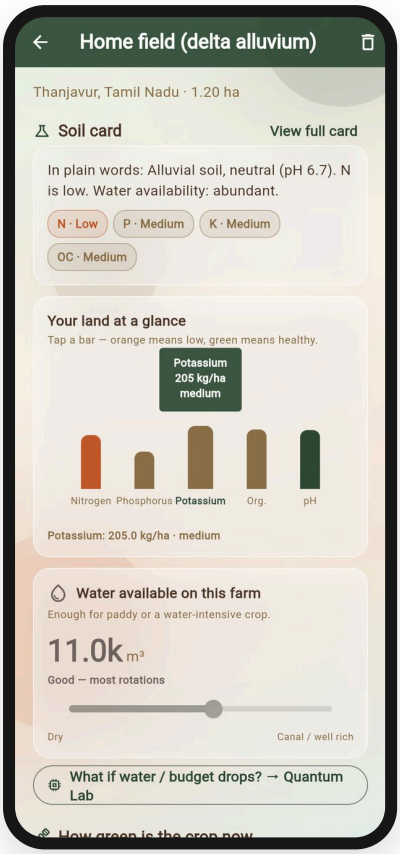
This is one of the best screens in the app: short, honest, and easy to act on.

Delete a farm

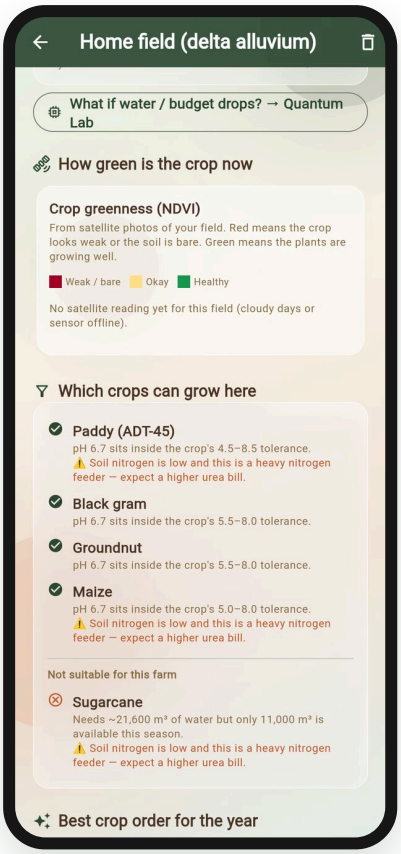
A bin icon on the farm page asks for confirmation before deleting, and says that the soil cards and plans will also go. This is good, safe design.

4.5 The farm page (the heart of the app)

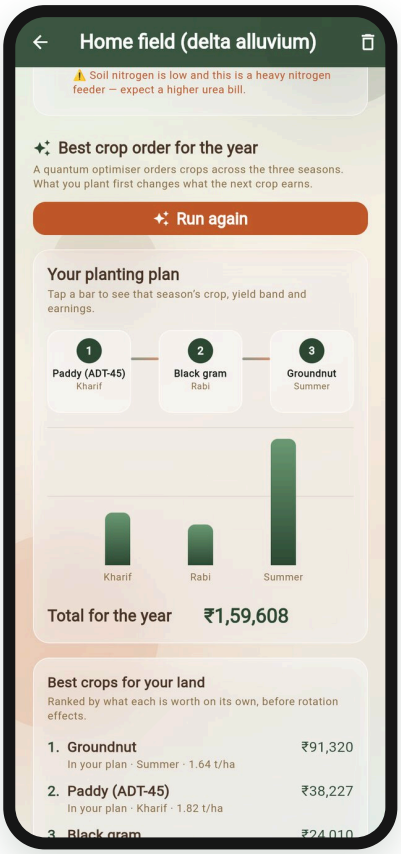
Tapping a farm opens one long page that tells the whole story of that farm, from top to bottom. These pictures follow the **Home field** demo farm (1.2 ha, alluvial soil).



1. Soil card + chart. Tap a bar to see the exact value.



2. Crop greenness + crop check.



3. Year plan. Season path and earnings chart.

Soil card and "Your land at a glance"

A summary in plain words, the nutrient chips, and a bar chart for nitrogen, phosphorus, potassium, organic carbon and pH. Orange bars are low, green bars are healthy. Below it, a read-only water card (11.0k m³ — "Good, most rotations") and a button that jumps to the Planner: *"What if water / budget drops?"*

How green is the crop now (NDVI)

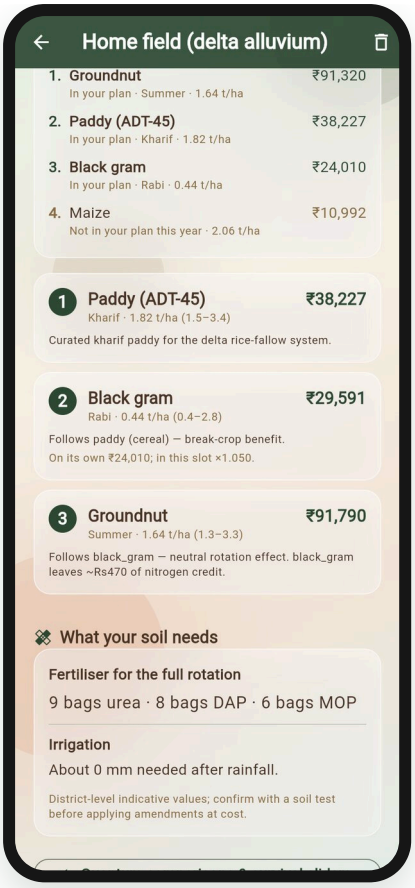
Meant to show a colour map of the field from satellite photos (red = weak, green = healthy) and a chart of the last few weeks. In our test it said *"No satellite reading yet"* because the Google Earth Engine key was not set up.

Which crops can grow here

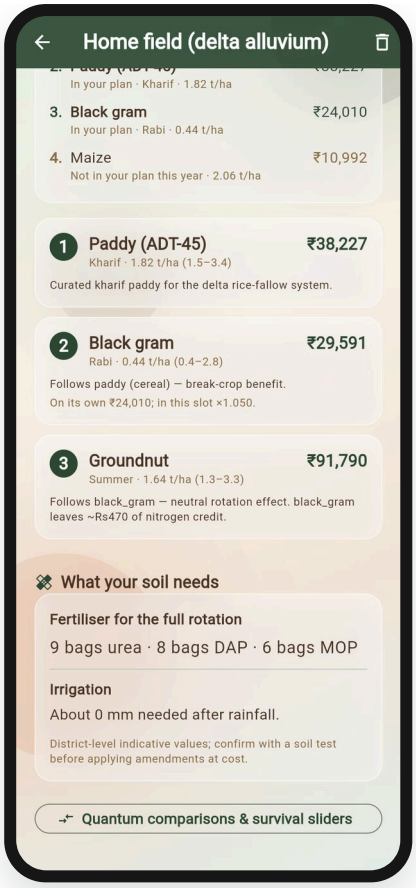
Green ticks for crops that pass, each with the deciding reason (for example "pH 6.7 sits inside the crop's 4.5–8.5 tolerance") and orange warnings like "soil nitrogen is low — expect a higher urea bill". Under **"Not suitable for this farm"**: Sugarcane — needs about 21,600 m³ of water but only 11,000 m³ is available.

Best crop order for the year

The big orange button **"Rank my crops"** (or "Run again") runs the optimiser. The result shows three season cards (Paddy → Black gram → Groundnut), a bar chart of money per season, and the **total for the year: ₹1,59,608**.



4. Crop ranking and season cards.



5. What your soil needs.

Best crops for your land

A second, different list: crops ranked by what each one earns **on its own**. Here groundnut is #1 (₹91,320). This is a smart touch — it explains why the most profitable crop is not always planted first. Crops left out of the plan say "Not in your plan this year".

Season cards ("why" each crop)

Each crop in the plan gets a card with its season, harvest guess and range (for example *Black gram · Rabi · 0.44 t/ha (0.4–2.8)*), its money, and one line explaining the choice: *"Follows paddy (cereal) — break-crop benefit."* It also shows the rotation effect, such as "on its own ₹24,010; in this slot ×1.050".

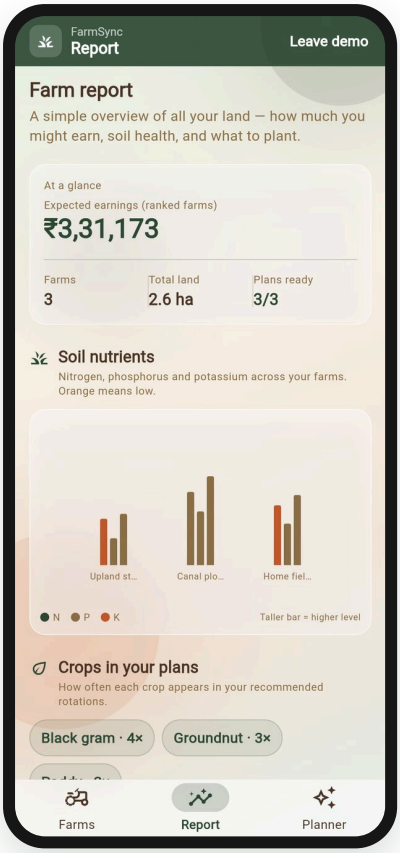
What your soil needs

One simple line a farmer can take to the shop: **9 bags urea · 8 bags DAP · 6 bags MOP** for the full rotation. If the soil is too acidic or too alkaline, it adds a soil fix (for example lime, with tonnes per hectare). It also gives irrigation need after rainfall ("About 0 mm needed after rainfall").

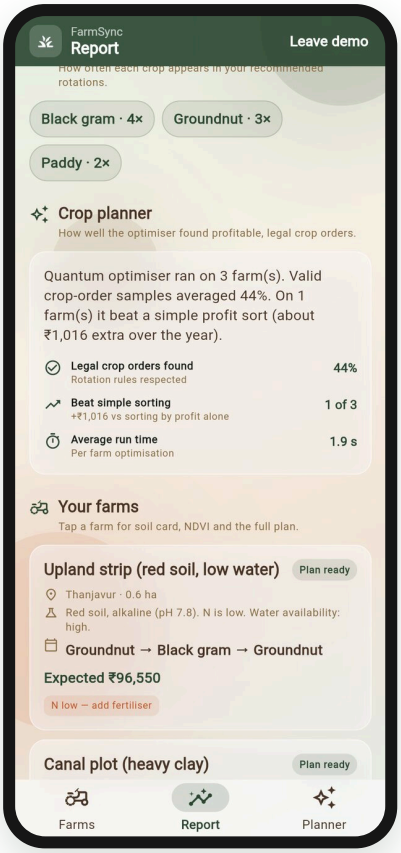
A final button **"Quantum comparisons & survival sliders"** jumps to the Planner tab with this farm already chosen.

Good: every number has a reason next to it. **Could be better:** the page is very long (about 5 phone screens). Tabs or folding sections would help.

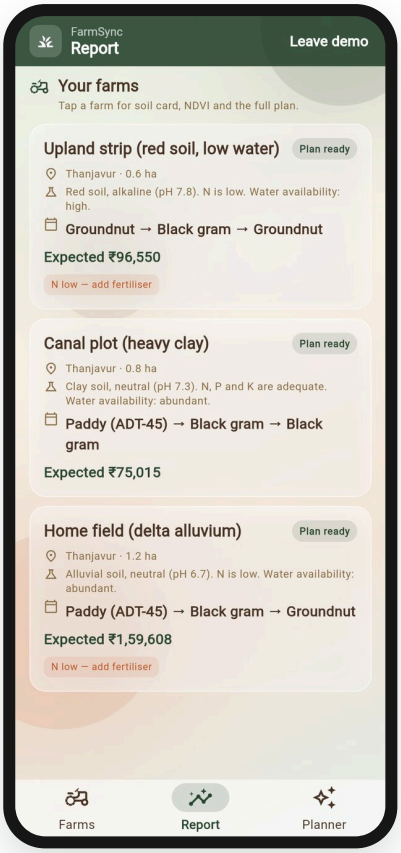
4.6 Report tab (all farms together)



At a glance + soil chart.



Crops in plans + optimiser summary.



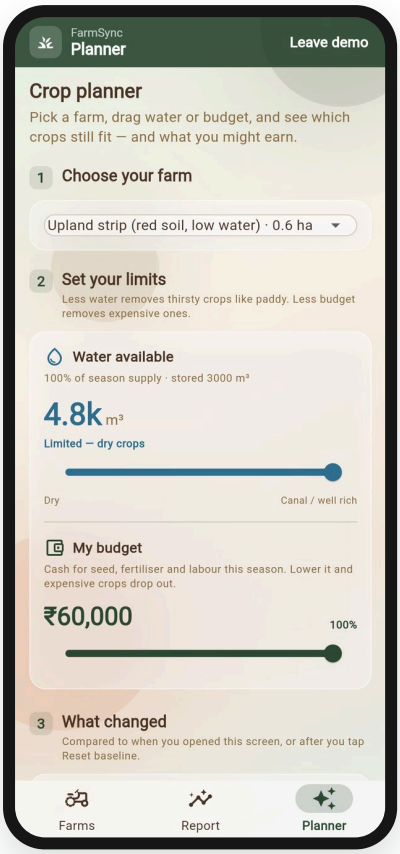
Your farms with status badges.

- **At a glance:** expected earnings for all ranked farms in big green numbers — **₹3,31,173** — then Farms (3), Total land (2.6 ha) and Plans ready (3/3). If some farms still need a plan, an orange note says how many.
- **Soil nutrients:** a bar chart with N, P and K for each farm side by side.
- **Crops in your plans:** chips showing how often each crop appears (Black gram · 4×, Groundnut · 3×, Paddy · 2×).
- **Crop planner summary:** a plain sentence ("Quantum optimiser ran on 3 farm(s)...") and three numbers: legal crop orders found (44%), beat simple sorting (1 of 3, +₹1,016), and average run time (1.9 s).
- **Your farms:** one card per farm with a **"Plan ready"** or **"Needs plan"** badge, the soil summary, the crop order, expected money, and problem chips such as **"N low — add fertiliser"**.

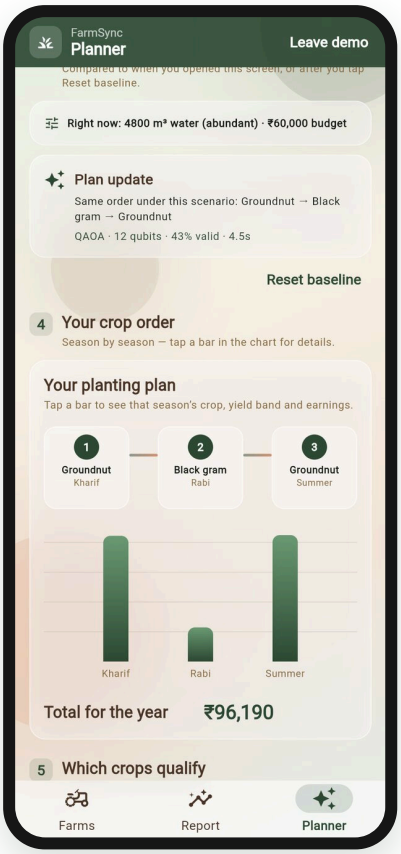
Confusing detail: the chart's legend says N = green, P = brown, K = orange. But the bars are coloured by Low / Medium / High, so an orange bar may be a *low nitrogen* bar, not potassium. Also, if the server cannot be reached, this tab shows the raw technical error text.

4.7 Planner tab ("what if" tool)

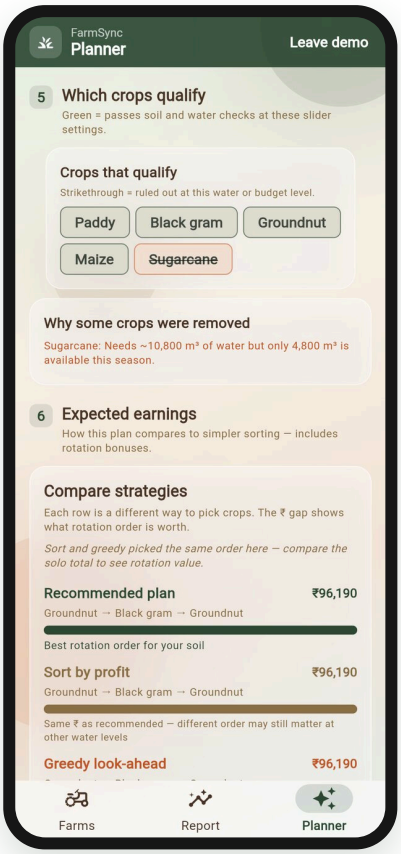
The Planner lets the farmer ask: *"What if I get less water, or have less money this season?"* It walks through six numbered steps.



Steps 1–2. Choose farm, set water and budget.



Steps 3–4. What changed, your crop order.

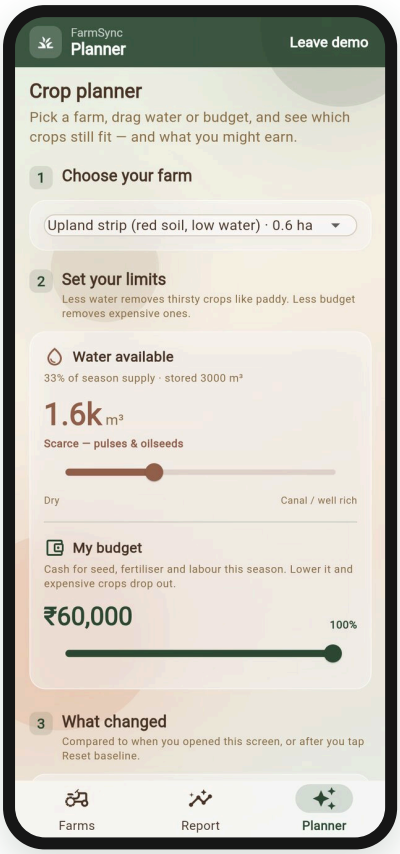


Steps 5–6. Crops that qualify, compare strategies.

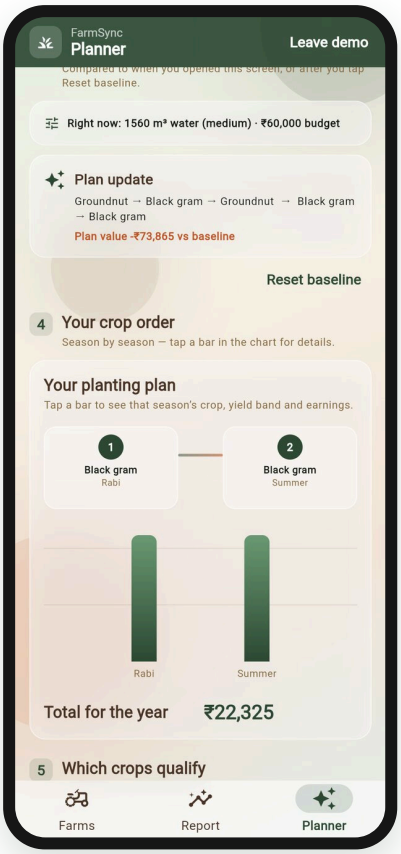
1. **Choose your farm** from a drop-down.
2. **Set your limits:** a water slider and a budget slider (₹). The plan re-runs about 0.6 seconds after you stop dragging.
3. **What changed:** "Right now: 4800 m³ water · ₹60,000 budget" and a "Plan update" note, with a "Reset baseline" button.
4. **Your crop order:** the same season path and chart as the farm page.
5. **Which crops qualify:** chips; crossed-out crops are ruled out, with the reason below.
6. **Expected earnings:** "Compare strategies" — the recommended plan vs sort-by-profit vs greedy look-ahead vs each crop alone. Here the recommended plan earns **₹37,051 more** than adding each crop's solo profit, which shows the value of good rotation.

4.8 The "what if" test: cutting the water

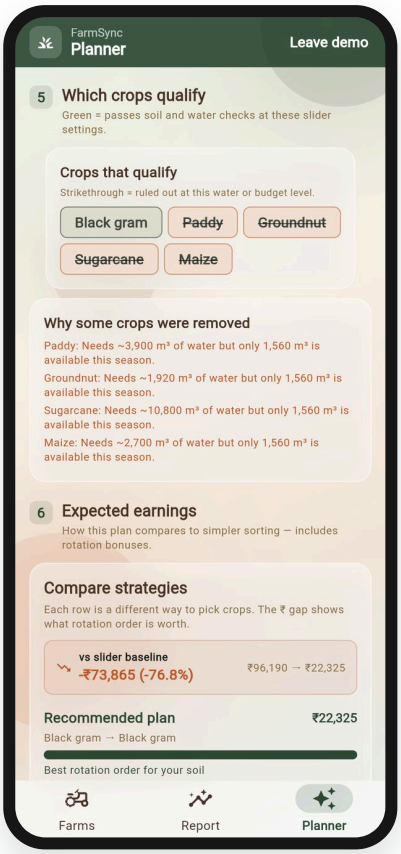
We dragged the water slider for the **Upland strip** farm down to about a third (1,560 m³). The plan updated by itself:



Water cut to 1.6k m³ — "Scarce — pulses & oilseeds".



Plan shrinks to Black gram → Black gram.



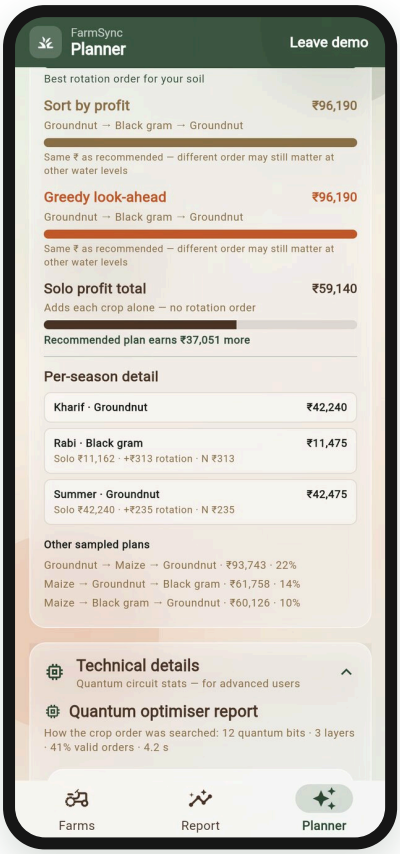
Reasons each crop dropped out.

- Paddy, groundnut, maize and sugarcane were all crossed out, each with a clear reason such as *"Paddy: needs ~3,900 m³ of water but only 1,560 m³ is available"*.
- The plan changed from **Groundnut → Black gram → Groundnut (₹96,190)** to **Black gram → Black gram (₹22,325)**. The app shows the drop plainly: **–₹73,865 (–76.8%)**.
- This makes the risk of a bad monsoon easy to see. It is a strong feature for teaching and for judges.

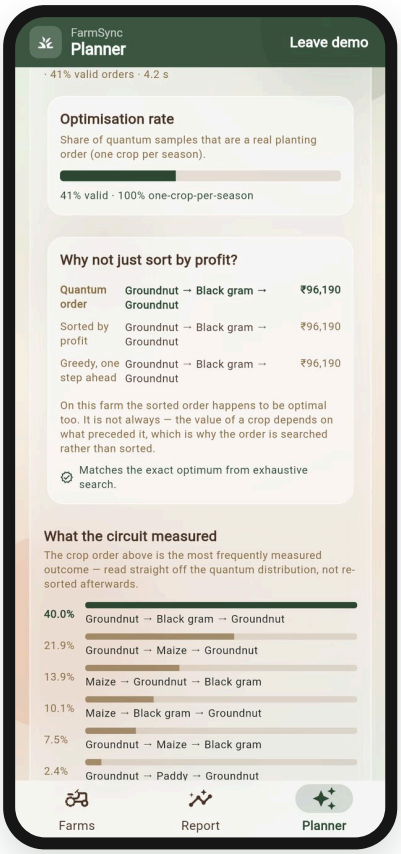
> **Problem found:** at 100% the slider starts at 4,800 m³, but this farm only has 3,000 m³ (the app uses "land size × 8,000" if that is bigger). So in the Planner, paddy and maize "qualify", while the farm page says paddy does not fit. The two water labels also disagree: the slider says "Limited — dry crops" while the line above says "abundant".

4.9 Technical details (for advanced users)

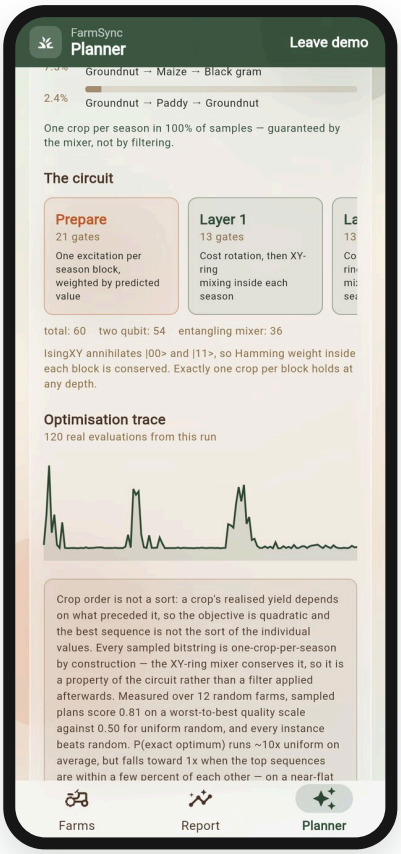
At the bottom of the Planner, a folded panel called **"Technical details — Quantum circuit stats"** opens the "Quantum optimiser report".



Opening the panel.



Why not just sort? + measured results.



The circuit + optimisation trace.

- **Summary line:** 12 quantum bits · 3 layers · 41% valid orders · 4.2 s.
- **Optimisation rate:** how many quantum samples were a real planting order.
- **Why not just sort by profit?** Quantum order vs sorted vs greedy, with money. It says plainly when sorting happens to give the same answer, and shows "Matches the exact optimum from exhaustive search".
- **What the circuit measured:** the top plans and how often each appeared (for example 40.0% for the winning order). The winner is read straight from these counts.
- **The circuit:** Prepare → Layer 1, 2, 3 → Measure, with gate counts, and a line chart of the 120 optimisation steps.

This panel is very good for judges and for showing that the quantum step is real. For a farmer, the text is too technical, so keeping it folded by default is the right choice.

5 Look and feel (UI design)

The design feels calm, natural and trustworthy — like soil, leaves and clay.

Colours

Deep green	Terracotta	Soil brown	Clay	Cream	Mist	Water blue
#2E4A34	#C1592A	#4A3527	#8B6F47	#F7F1E6	#E8F0E4	#2F6F8F
Top bar, main text, "good"	Main buttons, "low"	Body text	Hints, "medium"	Background	Background	Water slider

The colours carry meaning across the whole app: **orange = low / needs attention**, **brown = medium**, **green = healthy / good**. The water slider fades from orange (dry) to blue (wet). This is consistent and easy to learn.

Building blocks

- **Frosted "glass" cards:** see-through white panels with soft blur and rounded corners (18 pt), over a soft green-to-cream background with blurred colour blobs.
- **Large main text:** main text is 16–18 points, titles 22–30 points. Hints and notes are smaller (11–13 points).
- **Chips:** small rounded labels for soil levels and crops.
- **Big buttons:** main actions are wide terracotta buttons with an icon.
- **Charts you can tap:** soil bars and season bars show exact values when tapped.
- **Indian money format:** ₹1,59,608 (lakh style), not ₹159,608 — easier for Indian farmers to read.
- **Icons for every section:** flask for soil, satellite for greenness, filter for crop check, sparkle for the plan.

What works well

- Every number has a short reason in plain words next to it.
- Numbered steps on the Report and Planner tabs; helpful empty and error screens on the Farms tab.
- Safe actions: delete asks first; the soil card warns before spending money.

What could be better

- **Language:** English only. Tamil (and other local languages) would make a big difference. Voice read-out could help farmers who read less.
- **Long pages:** the farm page and the Planner are 5–8 screens long.
- **Some technical words** still show to farmers: "QAOA", "qubits", "NDVI", "LightGBM", "SPARQ" (sign-in badges and Planner notes).
- **Small grey hint text** (11–12 points on see-through cards) can be hard to read in sunlight, and the blur effect can be slow on cheap phones.
- **No dark mode** and no offline mode — farms often have weak mobile signal.

6 Behind the scenes

The app on the phone is the "face". A server does the thinking.

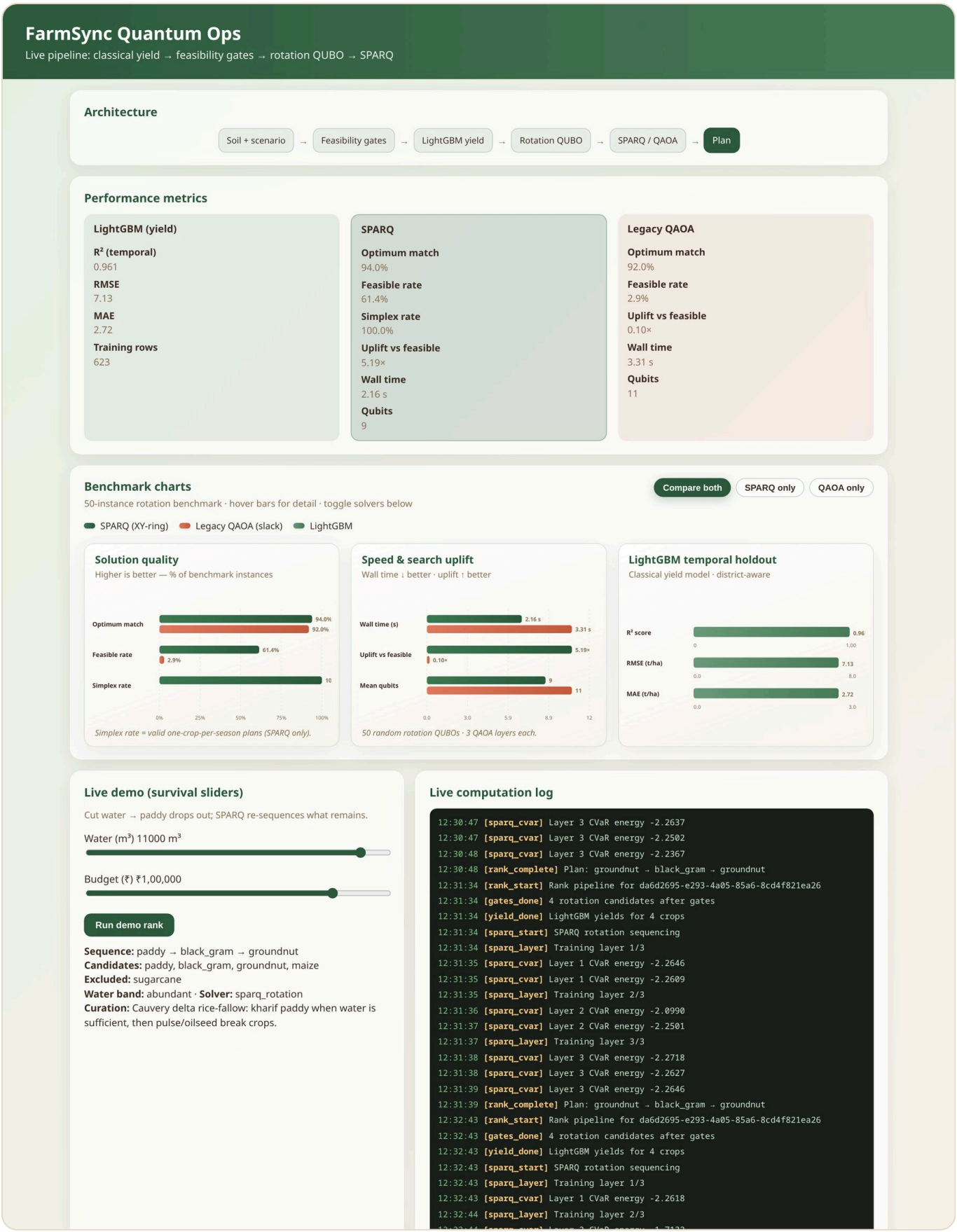
Part	What it uses	What it does
Phone app	Flutter (Dart), Riverpod	All the screens. Works on Android and iPhone from one code base.
Sign-in	Clerk + Google	Secure login; each farmer only sees their own farms.
Server	FastAPI (Python)	Receives requests from the app and runs the 5-step pipeline.
Database	SQLite (testing) / Postgres on Supabase (live)	Stores farmers, farms, soil cards and plans.
AI harvest model	LightGBM, 623 real records	Predicts tonnes per hectare plus a low–high range (P10–P90).
Quantum optimiser	PennyLane simulator ("SPARQ")	Finds the best crop order; compared live against older QAOA.
Card reader	Google ML Kit (on the phone)	Reads numbers from a photo of the Soil Health Card.
Weather	Open-Meteo, NASA POWER as backup	Rain, temperature and heat for the season.
Satellite	Sentinel-2 via Google Earth Engine	Crop greenness (NDVI). Optional — needs a key.
Reference data	Soil Health Card district data, crop rules, market prices	Fills gaps and turns harvests into rupees.

How accurate is the AI harvest model?

On past years it had never seen, it scored **$R^2 = 0.96$** (1.0 would be perfect). The project is honest that this number looks better than it is: it is only about 1% better than simply using the district average, and sugarcane's very large harvests make the score look high. In simple words: **good for comparing crops, not exact for one farm.**

The Ops page (web dashboard for the team)

The server also has a web page at **/ops** that shows the whole pipeline, performance numbers, benchmark charts, a live demo with sliders, and a live log of each step as it runs.



Real screenshot after pressing "Run demo rank". Benchmark: the new SPARQ method finds valid plans 61.4% of the time vs 2.9% for the older QAOA, uses fewer qubits (9 vs 11) and is faster (2.16 s vs 3.31 s). Small issues: the log box grows forever and repeats some lines, and the "Simplex rate" bar label is cut off ("10" instead of "100%").

7 Real results for the 3 demo farms

These numbers came from running the real engine on 23 October 2026, with live weather. Only the farmer's inputs were typed in.

Farm	Soil and water	Crops that can grow	Best order (Kharif → Rabi → Summer)	Money / year
Home field 1.2 ha	Alluvial, pH 6.7, N low 11,000 m³ water	4 of 5 Sugarcane out: needs ~21,600 m³	Paddy → Black gram → Groundnut	**₹1,59,608**
Canal plot 0.8 ha	Heavy clay, pH 7.3, N-P-K ok 9,000 m³ water	2 of 5 Groundnut, maize out: clay drains poorly. Sugarcane out: water	Paddy → Black gram → Black gram	**₹75,015**
Upland strip 0.6 ha	Red soil, pH 7.8, N low 3,000 m³ water	3 of 5 Paddy out: needs ~3,900 m³. Sugarcane out: water	Groundnut → Black gram → Groundnut	**₹96,550**

Soil advice given

Farm	Urea bags	DAP bags	MOP bags	Soil fix
Home field	9	8	6	None needed (pH is ideal)
Canal plot	5	5	3	None needed
Upland strip	1	4	4	Add organic matter (gypsum only if sodic soil is confirmed)

Quantum step on each farm

Farm	Qubits used	Valid plans in samples	Run time	Same as the best possible plan?
Home field	12	21%	3.5 s	Yes
Canal plot	6	63%	0.5 s	Yes
Upland strip	9	48%	1.7 s	Yes

One value worth a sanity check: paddy is predicted at about 1.8–1.9 t/ha on these farms, which looks low for the Cauvery delta. The model learns from district totals, not single farms, so check it against local yield records.

Why these numbers differ from the README

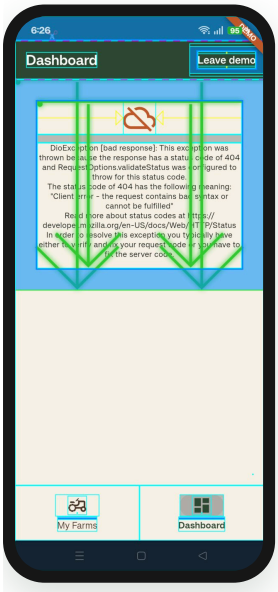
The README lists Home field as "groundnut ×3 — ₹2,67,028". Today the engine chose Paddy → Black gram → Groundnut for ₹1,59,608. This is expected: the engine uses **live weather**, so the harvest guesses change from day to day. The quantum step also uses random sampling, so the "other sampled plans" list changes a little each run, even when the top plan stays the same. It is worth telling users this, and updating the README table.

8 Testing and health check

Automatic tests are small programs that check the app still works after a change. We ran all of them.

What we checked	Result	What it means
Server tests (Python, pytest)	48 pass · 1 fail · 1 skip	Most of the engine is well tested (quantum rules, AI features, satellite maths).
Server API tests (tests/test_api.py)	Cannot run	A spacing mistake on line 392 (*IndentationError*) stops the whole file. Its 24 tests — sign-in, privacy between accounts, adding and ranking farms — are not running at all.
Quantum quality test	Failed 3 of 3 runs	Checks that the quantum plans are clearly better than random. Our runs scored 0.63 (needs more than 0.65), and once only 4 of 5 farms beat random. The documents claim 0.81 and "every instance". The randomness should be fixed (seeded) and the claim re-checked.
App tests (Flutter)	16 pass · 1 fail	The failing test still expects a cream background, but the app now uses a see-through background for the glass look. The test is out of date, not the app.
Code check (flutter analyze)	3 small notes	One unused setting and two style hints. Nothing serious.
Live server (rjs-production.up.railway.app)	Offline	Answers "Application not found" (error 404). The app points here by default, so installed apps cannot load any farms.
Local server (our test run)	Works	All screens loaded, ranking ran in 0.5–3.5 s per farm, the Ops page worked.

A real phone screenshot from the repo



flutter_01.png from the project folder (an older build, with debug lines on).

This picture was already in the project. It shows what a farmer sees when the server is missing: a cloud icon and a long technical message — *"DioException [bad response]… status code of 404…"*.

The Farms tab now turns this into a friendlier message, but the **Report tab still shows the raw error**. A farmer cannot understand or act on this text.

> **Fix:** bring the server back online, and show one friendly message everywhere, such as *"We can't reach FarmSync right now. Check your internet and try again."* with a Try again button.

9 Problems found

Listed from most to least important. "Where" points to the file in the project.

#	Problem (in simple words)	Priority	Suggested fix
1	**Live server is offline.** The default address returns "Application not found". Where: *flutter_app/lib/core/env.dart*	High	Redeploy the server, or point the app to a working one.
2	**Demo secret key is built into the app** and shown by "Show demo credentials". Anyone with the app can log into the demo account and add or delete its farms. Where: *env.dart*, *sign_in_page.dart*	High	Remove the default key from public builds. Make the demo account read-only.
3	**24 server tests never run** because of a spacing mistake. Where: *backend/tests/test_api.py:392*	High	Fix the indentation and re-run all tests.
4	**Quantum quality test fails** in our runs; the "0.81 vs 0.50" claim was not reproduced. Where: *backend/tests/test_quantum.py*	High	Fix the random seed, re-measure, and update the claim in PRD/README if needed.
5	**Planner starts with more water than the farm has** (4,800 vs 3,000 m³), so ruled-out crops come back. Water labels disagree ("Limited" vs "abundant"). Where: *quantum_lab_screen.dart*	Medium	Start the slider at the farm's real water. Use one rule for the water label.
6	**Sign-in box** says "Sign in to RJS" and is cut off at 200 points tall.	Medium	Rename the app in Clerk to FarmSync. Let the box grow to fit.
7	**English only**, while users are Tamil Nadu farmers.	Medium	Add Tamil (the server already has Tamil soil names).
8	**Raw error text** on the Report tab when the server fails.	Medium	Reuse the friendly error from the Farms tab.
9	**Crop greenness (satellite)** shows "No reading" unless Google Earth Engine is set up.	Medium	Set up the key on the live server, or hide the panel when there is no data.
10	**Report chart legend** does not match bar colours.	Low	Label each bar N, P, K, and keep colour only for Low/Medium/High.
11	**Flutter theme test out of date** (expects cream, app uses transparent).	Low	Update the test.
12	**Ops page log** grows forever and repeats lines; one chart label is cut.	Low	Give the log a fixed height; remove duplicates.
13	**Long crop names** push the price against the text on farm cards.	Low	Put the price on its own line or add spacing.
14	**README numbers are out of date** (live weather changes results). Test counts in README (55 / 12) also differ from today (74 / 17).	Low	Update README and explain that results follow the weather.

Note: making a web version of the app needs two extra fixes (a passkey script and file storage for the login service). This only matters if a web version is planned; the app is built for phones.

10 What to do next

A simple plan in three steps.

Now (this week)

- Bring the server back online.
- Remove the built-in demo key from public builds.
- Fix test_api.py line 392 and run all tests.
- Fix the quantum test's randomness and re-check the claim.
- Rename "RJS" to FarmSync in Clerk.

Next (this month)

- Start the Planner water slider at the farm's real water.
- One friendly error message everywhere.
- Tamil language.
- Set up Earth Engine for crop greenness, or hide it.
- Fix the chart legend and small layout issues.
- Update the README numbers.

Later (ideas)

- Offline mode: keep the last plan on the phone.
- Voice read-out of the plan.
- Shorter farm page with tabs or folding parts.
- More crops and more districts.
- Check yield guesses against real local records.
- Reminders for fertiliser dates (the server already has a schedule).

Final thoughts

FarmSync is a thoughtful, well-designed app with a clear purpose. The screens are clean and friendly, every result comes with a reason, and the quantum step is used for a real job (choosing the crop order) instead of just for show. The project documents are unusually honest about limits.

The main gaps are not in the idea or the design. They are in **operations and safety**: the live server is down, a demo key ships inside the app, and part of the test suite is not running. Fix those, add Tamil, and FarmSync will be ready for real farmers to try.

11 Word list (glossary)

Soil Health Card	A government card that lists the nutrients and pH of a farmer's soil.
N, P, K	Nitrogen, Phosphorus, Potassium — the three main plant foods. Measured in kg per hectare.
pH	How acidic (below 7) or alkaline (above 7) the soil is. Most crops like 6–7.5.
Organic carbon (OC)	How much natural organic matter is in the soil. More is usually better.
EC / salinity	How salty the soil is, in dS/m. High salt harms most crops.
Hectare (ha)	A unit of land: 100 m × 100 m, about 2.5 acres.
m³ (cubic metre)	A unit of water: 1,000 litres.
t/ha	Tonnes of harvest per hectare.
Kharif, Rabi, Summer	The three crop seasons: monsoon, winter, and hot dry season.
Crop rotation	Planting different crops one after another on the same land to keep the soil healthy.
Urea, DAP, MOP	Common fertilisers: urea gives nitrogen, DAP gives phosphorus (and some nitrogen), MOP gives potassium.
NDVI	A "greenness" number from satellite photos. Higher means healthier plants.
Feasibility gates	The yes/no rules that remove crops which cannot grow on a farm.
LightGBM	A type of AI model that learns patterns from past data. Used here to guess harvests.
P10 – P90 range	A likely range for the harvest: low-end and high-end guesses.
Qubit	The basic unit of a quantum computer, like a "bit" in a normal computer.
QAOA / SPARQ	Quantum methods for finding the best choice among many. SPARQ is this project's improved version.
Simulator	A normal computer pretending to be a quantum computer. FarmSync uses one.
Server / API	The computer program on the internet that the app talks to.
Clerk	The login service that handles "Sign in with Google".
Ops page	A web page for the team that shows how the engine is running.

Appendix A — Notes on how the pictures were taken

- The app was built from this project's code with Flutter 3.47.7 and opened in Google Chrome at phone size (412 × 915 points). The app normally runs on Android; we used the web build only to take pictures.
- To make the web build start, two small web-only patches were added in a temporary copy (a stub for the passkey script, and in-memory storage for the login box). The project files were not changed.
- The server ran on this computer with Python 3.14 and the project's pinned packages. Demo farms were created with *scripts/seed_demo.py*. Live weather came from Open-Meteo. Google Earth Engine was not set up, so satellite greenness had no data.
- The Ops page screenshot was taken at desktop width (1360 points). The older phone screenshot is the file *flutter_app/flutter_01.png* from the project.